

Problem

A particle of mass m slides on the inside of a smooth cone of semi-vertical angle α , whose axis points vertically upwards. Obtain the Hamiltonian function, using the distance r from the vertex, and the azimuth angle φ as generalised co-ordinates. Show that stable circular motion is possible for any value of r , and determine the corresponding angular velocity, ω . Find the angle α if the frequency of small oscillations about this circular motion is also ω .

Solution

Lagrangian for m in spherical polars *

$$\begin{aligned} L &= \frac{1}{2}m\dot{\mathbf{r}}^2 - V \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\alpha}^2 + r^2\sin^2\alpha\dot{\varphi}^2) - V(r, \alpha, \varphi) \end{aligned}$$

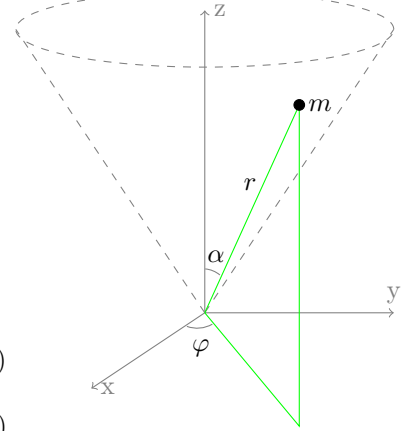
For motion inside cone α is constant, $\dot{\alpha} = 0$, hence

$$\begin{aligned} L &= \frac{1}{2}m(\dot{r}^2 + r^2\sin^2\alpha\dot{\varphi}^2) - V(r) \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\sin^2\alpha\dot{\varphi}^2) - mgr\cos\alpha \end{aligned}$$

The generalized momenta p_r and p_φ are given by

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} \quad \rightarrow \quad \dot{r} = \frac{p_r}{m} \quad (1)$$

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = mr^2\sin^2\alpha\dot{\varphi} \quad \rightarrow \quad \dot{\varphi} = \frac{p_\varphi}{mr^2\sin^2\alpha} \quad (2)$$



Hamiltonian for generalized co-ordinates r and φ

$$\begin{aligned} H(r, \varphi) &= (p_r\dot{r} + p_\varphi\dot{\varphi}) - L \quad (\text{or } T + V) \\ &= p_r\dot{r} + p_\varphi\dot{\varphi} - \frac{1}{2}m(\dot{r}^2 + r^2\sin^2\alpha\dot{\varphi}^2) + mgr\cos\alpha \\ &= m\dot{r}^2 + mr^2\sin^2\alpha\dot{\varphi}^2 - \frac{1}{2}m(\dot{r}^2 + r^2\sin^2\alpha\dot{\varphi}^2) + mgr\cos\alpha \\ &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\sin^2\alpha\dot{\varphi}^2 + mgr\cos\alpha \\ &= \frac{p_r^2}{2m} + \frac{p_\varphi^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha \end{aligned}$$

The first pair of Hamilton's equations reproduce (1) & (2) and the second pair give

$$-\dot{p}_r = \frac{\partial H}{\partial r} = -m\ddot{r} = mr\sin^2\alpha\dot{\varphi}^2 - mg\cos\alpha \quad \rightarrow \quad \ddot{r} = g\cos\alpha - r\sin^2\alpha\dot{\varphi}^2 \quad (3)$$

$$-\dot{p}_\varphi = \frac{\partial H}{\partial \varphi} = 0 \quad \rightarrow \quad p_\varphi = J \quad \text{i.e. angular momentum is constant} \quad (4)$$

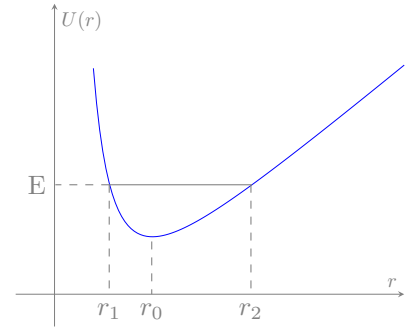
Thus the Hamiltonian may be written

$$H = \frac{1}{2}m\dot{r}^2 + \frac{J^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha = \frac{1}{2}m\dot{r}^2 + U(r)$$

Where $U(r) = \frac{J^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha$ is the 'effective potential energy'.

The total energy E (constant), is given by the Hamiltonian, hence if $E = U(r_0)$ then $\dot{r} = 0$ i.e. stable circular motion. This is the minimum point on the curve, hence if

$$\begin{aligned} dU/dr &= 0 \\ -\frac{J^2}{mr^3 \sin^2 \alpha} + mg \cos \alpha &= 0 \\ J^2 &= (mr^2 \sin^2 \alpha \dot{\phi})^2 = m^2 g r^3 \sin^2 \alpha \cos \alpha \\ \omega^2 = \dot{\phi}^2 &= \frac{g \cos \alpha}{r \sin^2 \alpha} \end{aligned}$$



If $E > U(r_0)$ the particle oscillates about $r_0 = g \cos \alpha / \omega^2 \sin^2 \alpha$ within the limits r_1 and r_2 .

For small oscillations near the minimum; $U(r)$ may be approximated by the Taylor series

$$\begin{aligned} U(r) &= \frac{1}{2} k r^2 \quad \text{where} \quad k = U''(r_0) \quad \text{hence} \\ \frac{d^2 U}{dr^2} &= \frac{3J^2}{mr^4 \sin^2 \alpha} \\ k = U''(r_0) &= \frac{3m^2 g \sin^2 \alpha \cos \alpha}{m \sin^2 \alpha} \frac{\omega^2 \sin^2 \alpha}{g \cos \alpha} \\ k &= 3m\omega^2 \sin^2 \alpha \end{aligned}$$

The equation of motion for small oscillations is simple harmonic

$$\ddot{r} + \omega'^2 r = 0 \quad \text{where} \quad \omega'^2 = \frac{k}{m}$$

Note: This is an approximate form of (3) for small oscillations.

If $\omega' = \omega$ then

$$\begin{aligned} \omega'^2 &= \omega^2 \\ 3\omega^2 \sin^2 \alpha &= \omega^2 \\ \sin \alpha &= 1/\sqrt{3} \\ \alpha &= \sin^{-1}(1/\sqrt{3}) \\ &= 35.3^\circ \end{aligned}$$

* See Appendix A *Co-ordinate Systems for Physics*